

Introduction to differential equations



Introduction to differential equations

- Definition of differential equations and their classification
- Solutions of differential equations
- Initial value problems
- Existence and uniqueness
- Methods of solution of first-order differential equations
- Mathematical models and examples

DIFFERENTIAL EQUATION:

- A Differential Equation is an equation containing the derivative of one or more dependent variables with respect to one or more independent variables.
- For Example,

$$\frac{dy}{dx} = 2xy$$

$$x \frac{dy}{dx} = y - 1$$

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial t^2} = 2 \frac{\partial u}{\partial t}$$

TYPES OF DIFFERENTIAL EQUATION:

ODE (ORDINARY DIFFERENTIAL EQUATION):

An equation contains only ordinary derivatives of one or more dependent variables of a single independent variable.

For Example,

$$dy/dx + 5y = e^x, \quad (dx/dt) + (dy/dt) = 2x + y$$

PDE (PARTIAL DIFFERENTIAL EQUATION):

An equation contains partial derivatives of one or more dependent variables of two or more independent variables.

For Example,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial t^2} = 2 \frac{\partial u}{\partial t}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Order of Differential Equation

The **order** of the differential equation is **order of the highest derivative** in the differential equation.

Differential Equation

ORDER

$$\frac{dy}{dx} = 2x + 3$$

1

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 9y = 0$$

2

$$\frac{d^3y}{dx^3} + \left(\frac{dy}{dx}\right)^4 + 6y = 3$$

3

Degree of Differential Equation

The **degree** of a differential equation is **power of the highest order derivative term** in the differential equation.

Differential Equation

Degree

$$\frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + ay = 0$$

1

$$\frac{d^3 y}{dx^3} + \left(\frac{dy}{dx} \right)^4 + 6y = 3$$

1

$$\left(\frac{d^2 y}{dx^2} \right)^3 + \left(\frac{dy}{dx} \right)^5 + 3 = 0$$

3

Linear Differential Equation

A differential equation is **linear**, if

1. **dependent variable and its derivatives are of degree one,**
2. **coefficients of a term does not depend upon dependent variable.**

Example: 1. $\frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + 9y = 0.$

is linear.

Example: 2. $\frac{d^3 y}{dx^3} + \left(\frac{dy}{dx} \right)^4 + 6y = 3$

is non - linear because in **2nd term** is not of degree one.

Example: 3. $x^2 \frac{d^2 y}{dx^2} + y \frac{dy}{dx} = x^3$

is non - linear because in 2nd term coefficient depends on y.

Example: 4. $\frac{dy}{dx} = \sin y$

is non - linear because $\sin y = y - \frac{y^3}{3!} + \dots$ is non - linear

nth – order linear differential equation

1. nth – order linear differential equation with constant coefficients.

$$P_0 \frac{d^n y}{dx^n} + P_1 \frac{d^{n-1} y}{dx^{n-1}} + P_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + P_n y = Q$$

If $P_0, P_1, P_2, \dots, P_n$ are all constants and Q is some function of x , then the equation is a linear differential equation with constant coefficients.

2. nth – order linear differential equation with variable coefficients

$$P_0(x) \frac{d^n y}{dx^n} + P_1(x) \frac{d^{n-1} y}{dx^{n-1}} + P_2(x) \frac{d^{n-2} y}{dx^{n-2}} + \dots + P_n(x) y = Q(x)$$

DEFINITION 1.1.2 Solution of an ODE

Any function ϕ , defined on an interval I and possessing at least n derivatives that are continuous on I , which when substituted into an n th-order ordinary differential equation reduces the equation to an identity, is said to be a **solution** of the equation on the interval.

INTERVAL OF DEFINITION You cannot think *solution* of an ordinary differential equation without simultaneously thinking *interval*. The interval I in Definition 1.1.2 is variously called the **interval of definition**, the **interval of existence**, the **interval of validity**, or the **domain of the solution** and can be an open interval (a, b) , a closed interval $[a, b]$, an infinite interval (a, ∞) , and so on.

EXAMPLE 1 Verification of a Solution

Verify that the indicated function is a solution of the given differential equation on the interval $(-\infty, \infty)$.

(a) $dy/dx = xy^{1/2}; \quad y = \frac{1}{16}x^4$ (b) $y'' - 2y' + y = 0; \quad y = xe^x$



SOLUTION One way of verifying that the given function is a solution is to see, after substituting, whether each side of the equation is the same for every x in the interval.

(a) From

$$\text{left-hand side:} \quad \frac{dy}{dx} = \frac{1}{16} (4 \cdot x^3) = \frac{1}{4} x^3,$$

$$\text{right-hand side:} \quad xy^{1/2} = x \cdot \left(\frac{1}{16} x^4 \right)^{1/2} = x \cdot \left(\frac{1}{4} x^2 \right) = \frac{1}{4} x^3,$$

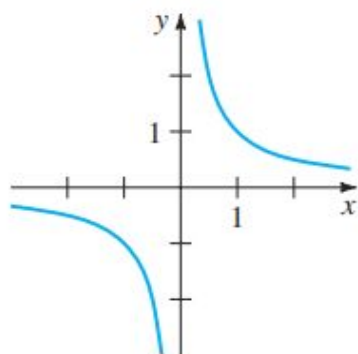
we see that each side of the equation is the same for every real number x . Note that $y^{1/2} = \frac{1}{4} x^2$ is, by definition, the nonnegative square root of $\frac{1}{16} x^4$.

(b) From the derivatives $y' = xe^x + e^x$ and $y'' = xe^x + 2e^x$ we have, for every real number x ,

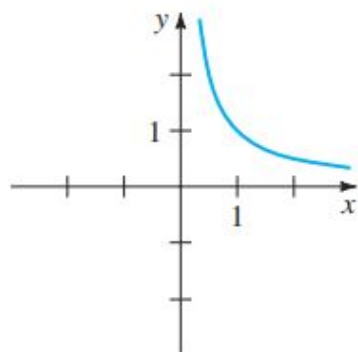
$$\text{left-hand side:} \quad y'' - 2y' + y = (xe^x + 2e^x) - 2(xe^x + e^x) + xe^x = 0,$$

$$\text{right-hand side:} \quad 0. \quad \blacksquare$$

Note, too, that in Example 1 each differential equation possesses the constant solution $y = 0$, $-\infty < x < \infty$. A solution of a differential equation that is identically zero on an interval I is said to be a **trivial solution**.



(a) function $y = 1/x$, $x \neq 0$



(b) solution $y = 1/x$, $(0, \infty)$

EXAMPLE 2 Function versus Solution

The domain of $y = 1/x$, considered simply as a *function*, is the set of all real numbers x except 0. When we graph $y = 1/x$, we plot points in the xy -plane corresponding to a judicious sampling of numbers taken from its domain. The rational function $y = 1/x$ is discontinuous at 0, and its graph, in a neighborhood of the origin, is given in Figure 1.1.1(a). The function $y = 1/x$ is not differentiable at $x = 0$, since the y -axis (whose equation is $x = 0$) is a vertical asymptote of the graph.

Now $y = 1/x$ is also a solution of the linear first-order differential equation $xy' + y = 0$. (Verify.) But when we say that $y = 1/x$ is a *solution* of this DE, we mean that it is a function defined on an interval I on which it is differentiable and satisfies the equation. In other words, $y = 1/x$ is a solution of the DE on *any* interval that does not contain 0, such as $(-3, -1)$, $(\frac{1}{2}, 10)$, $(-\infty, 0)$, or $(0, \infty)$. Because the solution curves defined by $y = 1/x$ for $-3 < x < -1$ and $\frac{1}{2} < x < 10$ are simply segments, or pieces, of the solution curves defined by $y = 1/x$ for $-\infty < x < 0$ and $0 < x < \infty$, respectively, it makes sense to take the interval I to be as large as possible. Thus we take I to be either $(-\infty, 0)$ or $(0, \infty)$. The solution curve on $(0, \infty)$ is shown in Figure 1.1.1(b). ■

Explicit solutions:

a solution in which the dependent variable is expressed solely in terms of the independent variable and constants.

Example:

$$y = \phi(x) = e^{0.1 x^2}$$

is an explicit solution of the ODE

$$\frac{dy}{dx} = 0.2xy$$

examples that $y = \frac{1}{16}x^4$, $y = xe^x$, and $y = 1/x$ are, in turn, explicit solutions of $dy/dx = xy^{1/2}$, $y'' - 2y' + y = 0$, and $xy' + y = 0$. Moreover, the trivial solution $y = 0$ is an explicit solution of all three equations.

Implicit solutions:

A relation $G(x, y) = 0$ is said to be an **implicit solution** of an ODE on an interval I provided there exists at least one function ϕ that satisfies the relation as well as the differential equation on I .

Example:

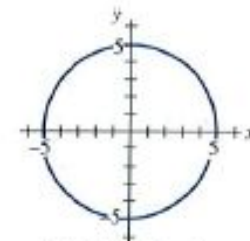
$$x^2 + y^2 = 25$$

is an implicit solution of the ODE

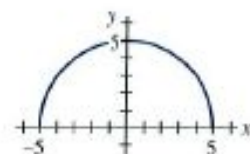
$$\frac{dy}{dx} = -\frac{x}{y}$$

on the interval $(-5, 5)$.

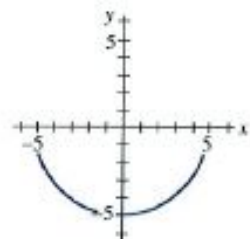
Notice that also $x^2 + y^2 - c = 0$ satisfies the ODE above.



(a) Implicit solution
 $x^2 + y^2 = 25$



(b) Explicit solution
 $y_1 = \sqrt{25 - x^2}, -5 < x < 5$



(c) Explicit solution
 $y_2 = -\sqrt{25 - x^2}, -5 < x < 5$

Families of solutions:

A solution ϕ of a first-order ODE $F(x, y, y') = 0$ can be referred to as an **integral** of the equation, and its graph is called an **integral curve**.

A solution containing an arbitrary constant (an integration constant) c represents a set

$$G(x, y, c) = 0$$

called a **one-parameter family of solutions**.

When solving an n th-order ODE $F(x, y, y', \dots, y^{(n)}) = 0$, we seek an **n -parameter family of solutions** $G(x, y, c_1, c_2, \dots, c_n) = 0$.

A single ODE can possess an infinite number of solutions!

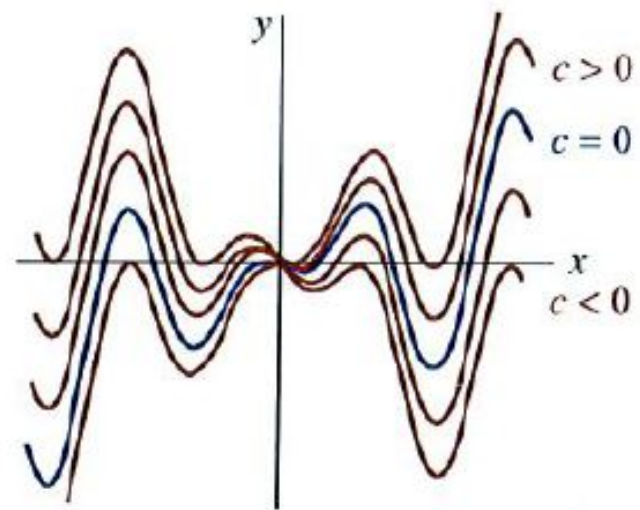
A particular solution:

is a solution of an ODE that is free of arbitrary parameters.

Example:

$y = cx - x \cos x$ is an explicit solution of $xy' - y = x^2 \sin x$ on $(-\infty, \infty)$.

The solution $y = -x \cos x$ is a particular solution corresponding to $c = 0$.



A singular solution:

a solution that can not be obtained by specializing any of the parameters in the family of solutions.

Example:

$y = (x^2/4 + c)^2$ is a one-parameter family of solutions of the DOE $dy/dx = xy^{1/2}$.

Also $y = 0$ is a solution of this ODE but it is not a member of the family above. It is a singular solution.

when $c = 0$

We get

$$y = x^4/16$$

In this case the trivial solution $y=0$ is a singular solution of the equation since it cannot be obtained from the family for any choice of the parameter c

The general solution:

If every solution of an n th-order ODE $F(x, y, y', \dots, y^{(n)}) = 0$ on an interval I can be obtained from an n -parameter family $G(x, y, c_1, c_2, \dots, c_n) = 0$ by appropriate choices of the parameters c_i , $i = 1, 2, \dots, n$ we then say that the family is the **general solution** of the differential equation.

Systems of differential equation:

A **system of ordinary differential equations** is two or more equations involving the derivatives of two or more unknown functions of a single independent variable.

Example:

$$\begin{aligned}\frac{dx}{dt} &= f(t, x, y) \\ \frac{dy}{dt} &= g(t, x, y)\end{aligned}$$

A **solution** of a system, such as above, is a pair of differentiable functions $x = \phi_1(t)$ and $y = \phi_2(t)$ defined on a common interval I that satisfy each equation of the system on this interval.

EXAMPLE 4 Using Different Symbols

The functions $x = c_1 \cos 4t$ and $x = c_2 \sin 4t$, where c_1 and c_2 are arbitrary constants or parameters, are both solutions of the linear differential equation

$$x'' + 16x = 0.$$

For $x = c_1 \cos 4t$ the first two derivatives with respect to t are $x' = -4c_1 \sin 4t$ and $x'' = -16c_1 \cos 4t$. Substituting x'' and x then gives

$$x'' + 16x = -16c_1 \cos 4t + 16(c_1 \cos 4t) = 0.$$

In like manner, for $x = c_2 \sin 4t$ we have $x'' = -16c_2 \sin 4t$, and so

$$x'' + 16x = -16c_2 \sin 4t + 16(c_2 \sin 4t) = 0.$$

Finally, it is straightforward to verify that the linear combination of solutions, or the two-parameter family $x = c_1 \cos 4t + c_2 \sin 4t$, is also a solution of the differential equation. ■

Initial value problem:

On some interval I containing x_0 , the problem of solving

$$\frac{d^n y}{dx^n} = f(x, y, y', \dots, y^{(n)})$$

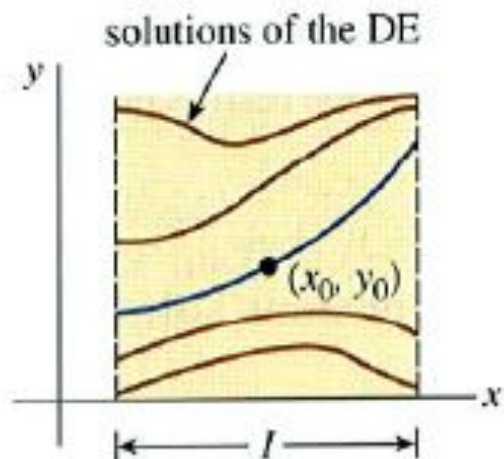
subject to the conditions

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \quad \dots, \quad y^{(n-1)}(x_0) = y_{n-1}$$

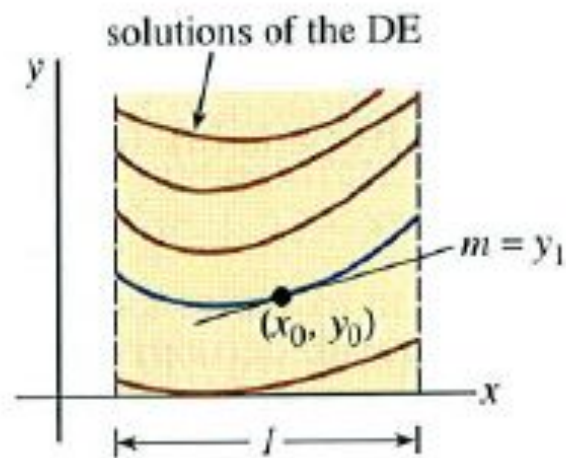
where $y_0, y_1, \dots, y_{n-1}$ are arbitrarily specified constants, is called an **initial value problem (IVP)**.

The conditions $y(x_0) = y_0, y'(x_0) = y_1, \dots, y^{(n-1)}(x_0) = y_{n-1}$ are called **initial conditions**.

First-order and Second-order IVPs:



$$\begin{aligned}\frac{dy}{dx} &= f(x, y) \\ y(x_0) &= y_0\end{aligned}$$



$$\begin{aligned}\frac{d^2y}{dx^2} &= f(x, y, y') \\ y(x_0) &= y_0 \\ y'(x_0) &= y_1\end{aligned}$$

Example:

$y = ce^x$ is a one-parameter family of solutions of the first order ODE $y' = y$ on the interval $(-\infty, \infty)$.

The initial condition $y(0) = 3$ determines the constant c :

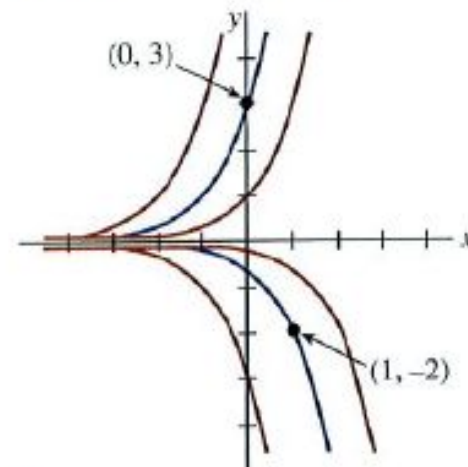
$$y(0) = 3 = ce^0 = c$$

Thus the function $y = 3e^x$ is a solution of the IVP defined by

$$y' = y, \quad y(0) = 3$$

Similarly, the initial condition $y(1) = -2$ will yield $c = -2e^{-1}$. The function $y = -2e^{x-1}$ is a solution of the IVP

$$y' = y, \quad y(1) = -2$$



Existence and uniqueness:

Does a solution of the problem exist? If a solution exist, is it unique?

Existence

Does the differential equation $dy/dx = f(x, y)$ possess solutions?

Do any of the solution curves pass through the point (x_0, y_0) ?

Uniqueness

When can we be certain that there is precisely one solution curve passing through the point (x_0, y_0) ?

Example: An IVP can have several solutions

Each of the functions

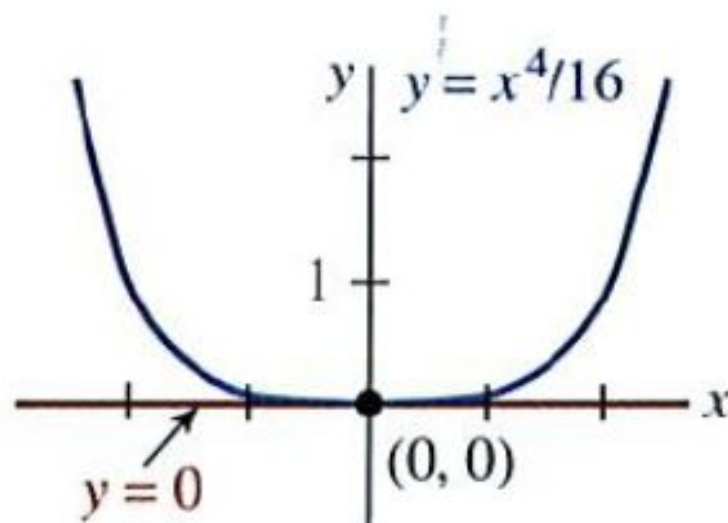
$$y = 0$$

$$y = x^4/16$$

satisfy the IVP

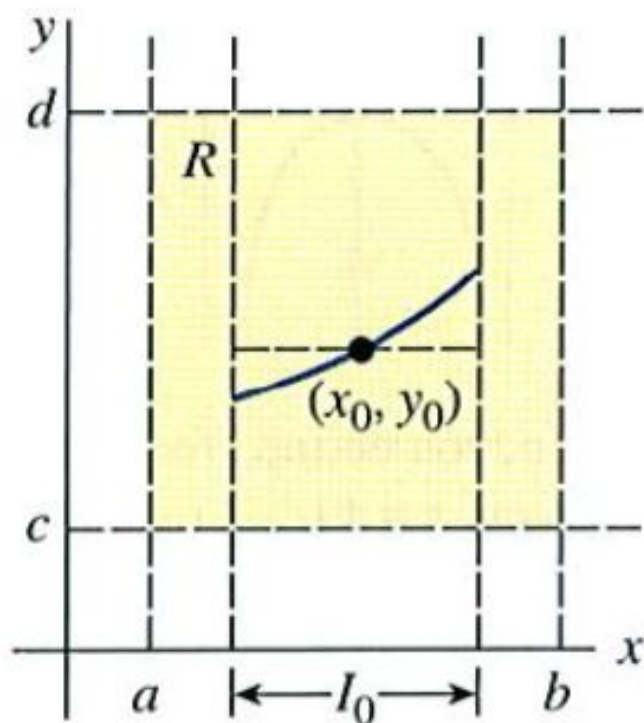
$$\frac{dy}{dx} = xy^{1/2}$$

$$y(0) = 0$$



Theorem: Existence of a unique solution

Let R be a rectangular region in the xy -plane defined by $a \leq x \leq b$, $c \leq y \leq d$, that contains the point (x_0, y_0) in its interior. If $f(x, y)$ and $\partial f / \partial y$ are continuous on R , then there exist some interval I_0 : $x_0 - h < x < x_0 + h$, $h > 0$, contained in $a \leq x \leq b$, and a unique function $y(x)$ defined on I_0 , that is a solution of the initial value problem (1).



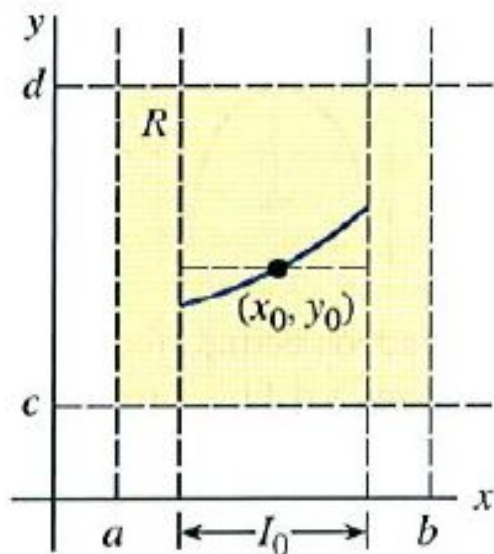
Distinguish the following three sets on the real x -axis:

the domain of the function $y(x)$;

the interval I over which the solution $y(x)$ is defined or exists;

the interval I_0 of existence AND uniqueness.

The theorem above gives no indication of the sizes of the intervals I and I_0 ; the number $h > 0$ that defines I_0 could be very small. Thus we should think that the solution $y(x)$ is *unique in a local sense*, that is near the point (x_0, y_0) .



Example: uniqueness

Consider again the ODE

$$\frac{dy}{dx} = xy^{1/2}$$

in the light of the theorem above. The functions

$$f(x, y) = xy^{1/2}$$
$$\frac{\partial f}{\partial y} = \frac{x}{2y^{1/2}}$$

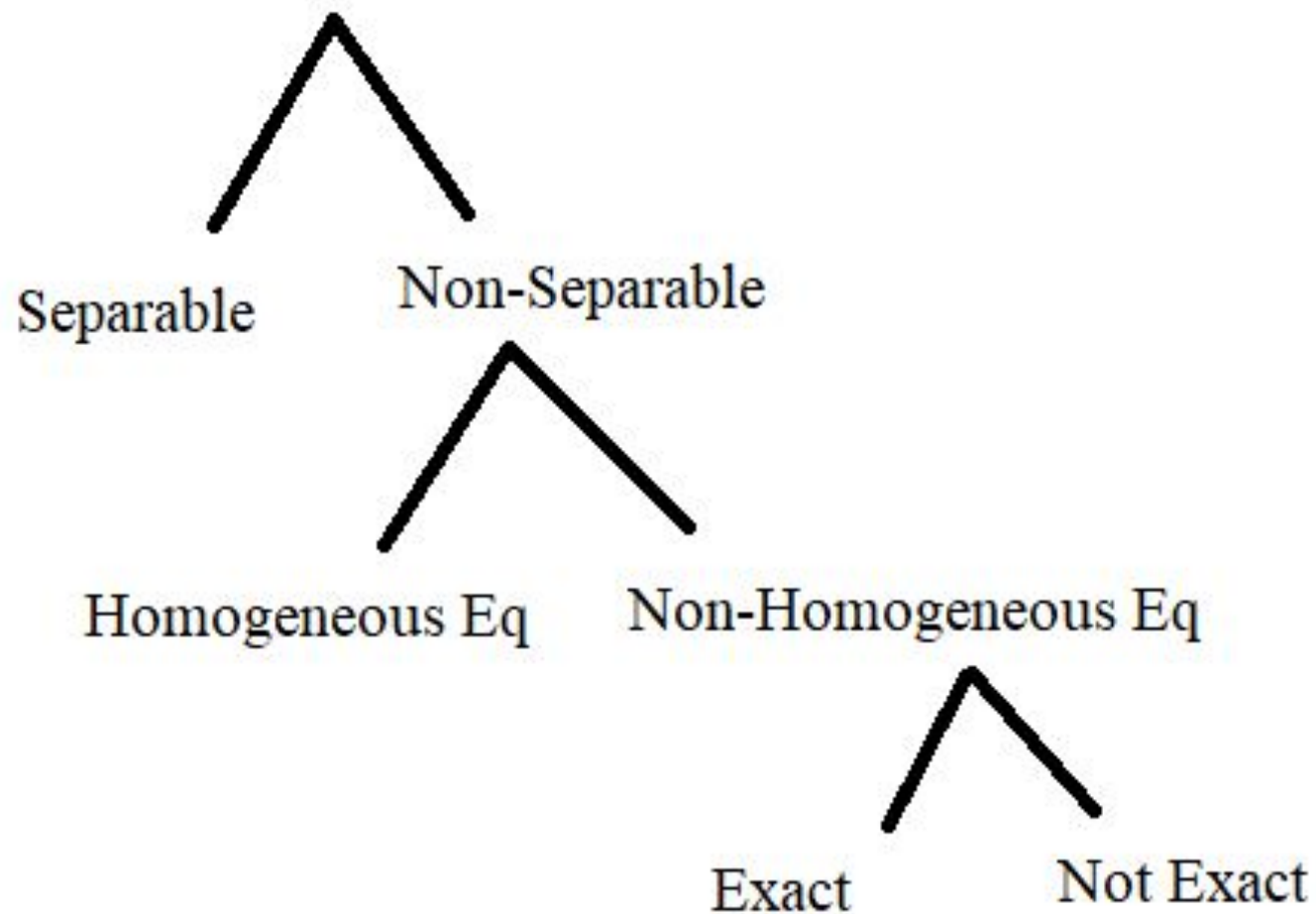
are continuous in the upper half-plane defined by $y > 0$.

The theorem allow us to conclude that through any point (x_0, y_0) , $y_0 > 0$, in the upper half-plane, there is an interval centered at x_0 , on which the ODE has a unique solution.

Remark

Since the second equation is not continuous on $y = 0$, the solutions of the DE is not unique in the xy -plane! The conditions in Theorem 1 are sufficient but not necessary. (That is, the statement is not “iff”.)

Classification of ODEs



Introduction and procedure

Separation of variables allows us to solve differential equations of the form

$$\frac{dy}{dx} = g(x)f(y)$$

The steps to solving such DEs are as follows:

1. Make the DE look like $\frac{dy}{dx} = g(x)f(y)$. This may be already done for you (in which case you can just identify the various parts), or you may have to do some algebra to get it into the correct form.
2. Separate the variables:
Get all the y 's on the LHS by multiplying both sides by $\frac{1}{f(y)}$ (i.e., dividing by $f(y)$):

$$\frac{1}{f(y)} \frac{dy}{dx} = g(x)$$

and get all the x 's on the RHS by 'multiplying' both sides by dx :

$$\frac{1}{f(y)} dy = g(x) dx$$

3. Integrate both sides:

$$\int \frac{1}{f(y)} dy = \int g(x) dx$$

This gives us an **implicit solution**.

4. Solve for y (if possible). This gives us an **explicit solution**.
5. If there is an initial condition, use it to solve for the unknown parameter in the solution function.
6. Check for any constant singular solutions. Remember: they must satisfy both the DE **and** the initial condition (if there is one).
7. If you have an initial condition, specify the interval of validity. If you don't have an initial condition, indicate the domain of the solution (even if it's not a single interval).

Example Solve the DE

$$y' = 5xy - 2x$$

subject to the initial condition $y(0) = 1$.

1. We could factor out just x from the RHS, but factoring out $5x$ leaves us with a coefficient of 1 on the y , which will make the integration a little easier later on:

$$\frac{dy}{dx} = \underbrace{5x}_{g(x)} \cdot \underbrace{\left(y - \frac{2}{5}\right)}_{f(y)}$$

2. Separate the variables:

$$\frac{1}{y - \frac{2}{5}} dy = 5x dx$$

3. Integrate both sides:

$$\begin{aligned} \int \frac{1}{y - \frac{2}{5}} dy &= \int 5x dx \\ \ln \left| y - \frac{2}{5} \right| &= \frac{5x^2}{2} + C_0 \quad (\text{an implicit solution}) \end{aligned}$$

4. Solve for y :

$$\begin{aligned} e^{\ln \left| y - \frac{2}{5} \right|} &= e^{\frac{5x^2}{2} + C_0} \\ \left| y - \frac{2}{5} \right| &= e^{\frac{5x^2}{2}} e^{C_0} = C_1 e^{\frac{5x^2}{2}} \\ \pm \left(y - \frac{2}{5} \right) &= C_1 e^{\frac{5x^2}{2}} \\ y - \frac{2}{5} &= \pm C_1 e^{\frac{5x^2}{2}} \\ y - \frac{2}{5} &= C e^{\frac{5x^2}{2}} \\ y &= C e^{\frac{5x^2}{2}} + \frac{2}{5} \end{aligned}$$

5. Apply the initial condition $y(0) = 1$ and solve for C:

$$1 = Ce^{\frac{5(0^2)}{2}} + \frac{2}{5}$$

$$1 = Ce^0 + \frac{2}{5}$$

$$1 = C + \frac{2}{5}$$

$$\frac{3}{5} = C$$

6. Check for constant singular solutions:

$$0 = 5xy - 2x = x(5y - 2)$$

So $y = \frac{2}{5}$ is a solution to the DE. However, it doesn't satisfy the initial condition, so $y = \frac{2}{5}$ is not a solution to the IVP.

7. The domain of the solution function is all real numbers, so our interval is $(-\infty, \infty)$.

So, the final solution is

$$y = \frac{3}{5}e^{\frac{5x^2}{2}} + \frac{2}{5},$$

$$I = (-\infty, \infty)$$



1. Population Growth and Decay (in statistics)

Problem: A population grows at the rate of 5% per year. How long does it takes for the population to double?

Solution: Let the initial population be p_0 and let the population after t years be p .

Then we get,

$$\begin{aligned}\frac{dp}{dt} &= \frac{5}{100}p \\ \Rightarrow \frac{dp}{p} &= \frac{5}{100} dt \\ \Rightarrow \frac{dp}{p} &= \frac{1}{20} dt \dots\dots\dots (1)\end{aligned}$$

which is a separable differential equation.

Now we integrating (1) and we get,

$$\Rightarrow \ln p = \frac{t}{20} + c \dots \dots \dots (2)$$

We put the initial value in (2) i.e. $t = 0$ and $p = p_0$

$$\ln p_0 = c$$

$$\Rightarrow c = \ln p_0 \dots \dots \dots (3)$$

We get from (2) and (3),

$$\Rightarrow \ln p = \frac{t}{20} + \ln p_0$$

$$\Rightarrow t = 20 \ln\left(\frac{p}{p_0}\right)$$

when, $p = 2p_0$ then,

$$\Rightarrow t = 20 \ln\left(\frac{2p_0}{p_0}\right)$$

$$= 20 \ln 2$$

$$= 13.86 \approx 14 \text{ years}$$

Hence the population is double in 14 years.

2. Determination of curves that have certain geometrical properties.

Problem: The slope of the tangent at a point (x, y) on a curve is $(-x/y)$.
If the curve passes through the point $(3, -4)$. Find the equation of the curve.

Solution: We know the slope of a curve at point (x, y) is $\frac{dy}{dx}$

Thus by the problem

$$\frac{dy}{dx} = \frac{-x}{y}$$

$$\Rightarrow ydy + xdx = 0 \dots\dots\dots (1)$$

Which is a separable differential equation.

Now integrating (1) and we get,

$$\frac{y^2}{2} + \frac{x^2}{2} = c$$
$$\Rightarrow x^2 + y^2 = 2c \dots \dots \dots (2)$$

where c is an arbitrary constant.

Since the curve passes through the point (3, -4)

thus from (2),

$$3^2 + (-4)^2 = 2c$$
$$\Rightarrow c = \frac{25}{2}$$

Hence,

$$x^2 + y^2 = 25,$$

which is the required equation of the curve.

Good Luck